Day 4 — The Equinoxes (Viṣuva)

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will define an equinox as the day when day and night are equal, identify the two equinoxes in the year, and explain — in terms of axial tilt — why the days are equal on those two dates.

Materials

- The Day 2 torch-globe setup
- A printable day-length graph (x-axis = months, y-axis = day length in hours; blank for the students to fill)
- Student notebooks

Vocabulary introduced today

- *viṣuva* (IAST: *viṣuva*; lit. "equal, balanced") — an **equinox**, one of the two days a year when day and night have equal length everywhere on Earth (~12 hours each). Two of them: the **vernal/March equinox** (~21 March) and the **autumnal/September equinox** (~23 September).
- *viṣuvad-dina* — another classical name, lit. "the equal day."

Lesson flow

Warm-up (5 min)

- Prompt: *"What does 'equinox' literally mean?"* — break it down: equi (equal) + nox (night). Latin for "equal night." The Sanskrit *viṣuva* means the same.
- Recall yesterday's diagram. Today's focus is on the two SIDEWAYS positions of Earth (March and September).

Core (20 min)

1. **The geometry of an equinox.** Set up the torch-globe again:
    - Tilt the globe so the N pole leans neither toward nor away from the torch — it's "sideways."
    - Notice the **terminator** (the line between light and dark on the globe) — it now passes through both poles, dividing Earth into two equal hemispheres of day and night.
    - **Consequence:** every place on Earth gets 12 hours of day and 12 hours of night.
2. **Why "equal" everywhere?** The day length at any latitude depends on what fraction of that latitude-line is in daylight. On the equinox, the day/night terminator passes through both poles — so every latitude-line is exactly half lit, half dark.

3. **Two equinoxes per year:**

Date	Name	What happens after
~21 March	**Vernal / spring equinox** (*vasanta-viṣuva*)	Days get longer in N hemisphere → spring → summer
~23 September	**Autumnal / autumn equinox** (*śarad-viṣuva*)	Days get shorter in N hemisphere → autumn → winter

4. **Two equinoxes + two solstices = four turning points of the year.** Draw on board:

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           ~21 June (summer solstice)
                 ☀ Sun highest at noon, longest day
                    (turning point — uttarāyaṇa ends, dakṣiṇāyana begins)
                    /
     ~23 Sept (autumn equinox / śarad-viṣuva)
           day = night everywhere
                    /
           ~21 Dec (winter solstice)
                 ☀ Sun lowest at noon, shortest day
                    (turning point — dakṣiṇāyana ends, uttarāyaṇa begins)
                    /
     ~21 Mar (spring equinox / vasanta-viṣuva)
           day = night everywhere
```

5. **A note on the dates:** The dates shift by ±1 day depending on which year (leap years, the slight unevenness of Earth's orbit). Modern almanacs give exact dates and times.

Activity (15 min) — The day-length graph

In pairs, students fill a blank graph:

- x-axis: months January to December
- y-axis: day length in hours (8 to 14)

The teacher provides a small table of approximate day lengths in Delhi (latitude 28.6° N):

Month	Day length (Delhi)
Jan	~10 h 25 min
Feb	~11 h 05 min
Mar	~12 h 00 min
Apr	~12 h 50 min
May	~13 h 30 min

Month	Day length (Delhi)
Jun	~13 h 50 min
Jul	~13 h 40 min
Aug	~13 h 00 min
Sep	~12 h 10 min
Oct	~11 h 25 min
Nov	~10 h 40 min
Dec	~10 h 15 min

Students plot the points, connect them, and mark:

- The **two highest points** (close together in June) — summer solstice.
- The **two lowest points** (close together in December) — winter solstice.
- The **two crossings of the 12-hour line** — the equinoxes.

Discussion (last 3 min):

- Why is the difference between longest and shortest day only ~3.5 hours in Delhi, but ~16 hours in London (latitude 51° N)? *(Answer: the higher the latitude, the more extreme the day-length swing.)*
- What would this graph look like at the equator? *(Answer: nearly flat at 12 h all year.)*
- At the North Pole? *(Answer: 24 h of daylight in summer, 24 h of darkness in winter.)*

Wrap-up (5 min)

- Exit ticket: *"Why are equinoxes called 'equal'? Answer in one sentence."*
- Preview Day 5: *"Tomorrow — the 12 rāśis. The Sun moves through one each month."*

Student-facing content

> **Equinox = day equals night.**

Sanskrit	Date	Position of Earth	Day length
vasanta-viṣuva	~21 March	tilt sideways	12 h day, 12 h night
śarad-viṣuva	~23 September	tilt sideways the other way	12 h day, 12 h night

> Together with the two **solstices** (June and December), these are the four turning-points of the year.
>
> The Sun crosses the celestial equator on each equinox — moving from the Southern celestial hemisphere into the Northern (in March), or vice versa (in September).

Homework

- Compare the sunrise / sunset times in your local newspaper or weather app TODAY with the times exactly six months from now (you can look up future-six-months online or in a printed almanac). Calculate the difference in day length. One paragraph.

Differentiation

- **One tier up:** What is the "true" length of the solar year? Why is it 365.25 days and not 365 exactly? How does the calendar handle this? (Answer involves leap years; the Gregorian system; the Indian *adhikamāsa* in lunisolar context.)
- **One tier down:** Just remember: two equinoxes, two solstices, four turning-points. Mark all four on a calendar.

Teacher notes

- Some students will object: "the newspaper says sunrise was at 6:23, sunset at 18:31 — that's 12 hours and 8 minutes, not 12 hours exactly." That's because the Sun is a **disc** (not a point) and "sunrise" is when the **top** of the disc clears the horizon, and "sunset" is when the **last** of the disc disappears. So there's a small offset (~5–10 min depending on latitude). Also, atmospheric refraction bends the Sun's light, making it visible slightly before geometric sunrise. Accept the observation and explain it briefly.
- The day-length graph activity often produces the "aha" moment — students *see* the year's shape.

Citation(s) used in this lesson

- Day-length figures are approximate values for Delhi (28.6° N), computable from standard spherical-astronomy formulas. Verifiable in any almanac.